Integrated Quality Management System

HEAVY CRANE RENTAL COMPANY

Connecting SIPOC · Fishbone · Feedback Loops · Flowchart · Optimization (Bottleneck Logic)

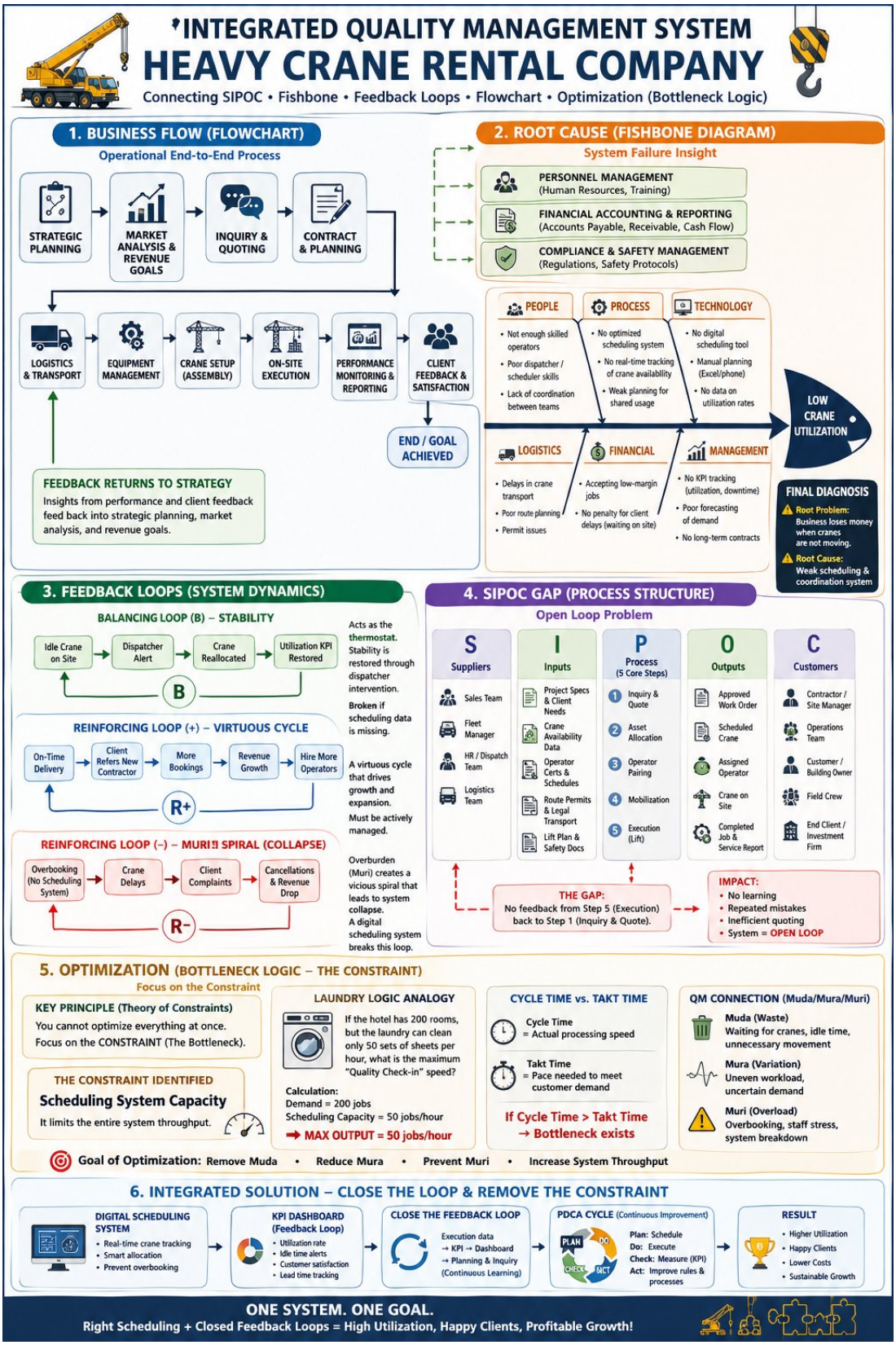

Figure 1 – Integrated QMS Infographic Overview

EXECUTIVE SUMMARY

This report presents a comprehensive Integrated Quality Management System (QMS) designed specifically for a Heavy Crane Rental Company. The framework synthesises six proven management and engineering tools — Business Flowchart (SIPOC-aligned), Root-Cause Fishbone Analysis, Feedback Loop System Dynamics, SIPOC Gap Analysis, Theory of Constraints (Bottleneck Logic), and an Integrated Improvement Solution — into a single cohesive model that diagnoses operational inefficiencies and prescribes actionable remedies.

Core Problem Identified: The business loses money when cranes remain idle. Root cause: a weak scheduling and coordination system that creates an open-loop process with no feedback from execution back to planning.

1. BUSINESS FLOW — Operational End-to-End Process

The operational lifecycle of the company follows eight sequential stages that transform a strategic objective into measurable client outcomes:

- **Strategic Planning:** Sets long-term direction, market targets, and revenue goals.
- **Market Analysis & Revenue Goals:** Identifies demand patterns, competitor positioning, and pricing benchmarks.
- **Inquiry & Quoting:** Captures client requirements and produces cost-accurate proposals.
- **Contract & Planning:** Formalises agreements and allocates crane resources per project.
- **Logistics & Transport:** Coordinates crane movement, route permits, and delivery scheduling.
- **Equipment Management:** Ensures crane availability, maintenance status, and certification compliance.
- **Crane Setup (Assembly):** On-site installation, safety checks, and operator briefing.
- **On-Site Execution & Performance Monitoring:** Real-time tracking of lift operations and KPI capture.
- **Client Feedback & Satisfaction:** Collects post-project data to feed back into strategic planning.

Key Insight: Client feedback and performance data must loop back to Strategic Planning to enable continuous improvement. Without this closed loop the system cannot self-correct.

2. ROOT CAUSE ANALYSIS — Fishbone (Ishikawa) Diagram

The Fishbone Diagram maps the causes of **Low Crane Utilisation** — the primary symptom that translates directly into revenue loss. Causes are grouped across six dimensions:

Dimension	Key Root Causes
People	Insufficient skilled operators; weak dispatcher/scheduler capability; poor cross-team coordination.
Process	No optimised scheduling system; no real-time crane-availability tracking; ad-hoc shared-usage planning.
Technology	No digital scheduling tool; manual planning via Excel/phone; no utilisation-rate data.
Logistics	Crane-transport delays; poor route planning; permit bottlenecks.
Financial	Acceptance of low-margin jobs; no penalty clauses for client delays; absence of long-term contracts.
Management	No KPI tracking for utilisation/downtime; poor demand forecasting; reactive rather than proactive culture.

Final Diagnosis: the business loses revenue whenever cranes sit idle due to a lack of real-time visibility and structured scheduling governance.

3. FEEDBACK LOOPS — System Dynamics

System Dynamics modelling reveals three behavioural archetypes that govern company performance:

B – Balancing Loop (Stability)

Idle Crane → Dispatcher Alert → Crane Reallocated → Utilisation KPI Restored. Acts as a thermostat. Effective only when the dispatcher has real-time data. Broken when scheduling data is absent or delayed.

R+ – Reinforcing Loop (Virtuous Cycle)

On-Time Delivery → Client Refers New Contractor → More Bookings → Revenue Growth → Hire More Operators → On-Time Delivery. A self-amplifying growth engine that must be actively managed to avoid over-expansion.

R– – Reinforcing Loop (Collapse / Muri Spiral)

Overbooking → Crane Delays → Client Complaints → Cancellations & Revenue Drop → Overbooking (pressure to compensate). A vicious cycle triggered by overbooking; a digital scheduling system is the circuit-breaker.

4. SIPOC GAP ANALYSIS — Open-Loop Problem

The SIPOC framework (Suppliers → Inputs → Process → Outputs → Customers) exposes a critical structural gap:

S – Suppliers	Sales Team, Fleet Manager, HR/Dispatch Team, Logistics Team
I – Inputs	Project Specs, Crane Availability Data, Operator Certs, Route Permits, Safety Docs
P – Process	5 Steps: ① Inquiry & Quote ② Asset Allocation ③ Operator Pairing ④ Mobilisation ⑤ Execution (Lift)
O – Outputs	Approved Work Order, Scheduled Crane, Assigned Operator, Crane on Site, Completed Job & Service Report
C – Customers	Contractor/Site Manager, Operations Team, Customer/Building Owner, Field Crew, End Client/Investment Firm

The Gap: There is NO feedback pathway from Step 5 (Execution) back to Step 1 (Inquiry & Quote). This open-loop structure produces repeated mistakes, inefficient quoting, and no organisational learning.

5. OPTIMISATION — Theory of Constraints (Bottleneck Logic)

Key Principle: You cannot optimise everything simultaneously. Focus exclusively on THE CONSTRAINT — the single resource or step that limits total system throughput.

The Constraint Identified: Scheduling System Capacity. The manual, fragmented scheduling process caps maximum output at approximately 50 jobs/hour regardless of how many cranes or operators are available.

> **Laundry Logic Analogy:** A 200-room hotel can prepare 200 rooms per shift, but if the laundry can only clean 50 sets of sheets per hour, the maximum 'Quality Check-in' speed equals 50 rooms/hour — irrespective of front-desk or housekeeping capacity. The laundry IS the bottleneck.

Cycle Time vs. Takt Time:

- **Cycle Time** = actual processing speed of the scheduling system (manual = slow).
- **Takt Time** = pace required to meet customer demand (200 jobs ÷ scheduling capacity).
- **If Cycle Time > Takt Time → Bottleneck exists → Revenue is lost.**

QM Connection — Muda / Mura / Muri:

Waste Type	Definition	Manifestation in Crane Rental
Muda (Waste)	Non-value-adding activities	Waiting for cranes, idle time, unnecessary crane movement
Mura (Variation)	Uneven workload / demand swings	Uneven booking load, uncertain demand cycles
Muri (Overload)	Pushing beyond safe capacity	Overbooking, staff stress, equipment breakdown

Goal of Optimisation: Remove Muda · Reduce Mura · Prevent Muri · Increase System Throughput

6. INTEGRATED SOLUTION — Close the Loop & Remove the Constraint

The integrated solution combines four interdependent components into one system:

■ **Digital Scheduling System:** Real-time crane tracking; smart allocation algorithms; prevention of overbooking. Directly attacks the identified constraint and converts the open loop into a closed loop.

■ **KPI Dashboard (Feedback Loop):** Live monitoring of utilisation rate, idle-time alerts, customer satisfaction scores, and lead-time tracking. Provides the data signal that drives the Balancing Loop.

■ **Close the Feedback Loop:** Execution data flows automatically to the KPI Dashboard, then back to Planning & Inquiry — enabling continuous organisational learning and quoting accuracy.

■ **PDCA Cycle (Continuous Improvement):** Plan (schedule jobs) → Do (execute lifts) → Check (measure KPIs) → Act (improve rules and processes). Embeds a systematic improvement cadence.

Expected Outcome	Business Impact
Higher crane utilisation rate	Direct revenue increase from previously idle assets
Happier clients	Repeat business and referral-driven growth (R+ loop)
Lower operational costs	Elimination of waste, rework, and emergency logistics
Sustainable growth trajectory	Scalable operations backed by data-driven decisions

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ONE SYSTEM. ONE GOAL.

Right Scheduling + Closed Feedback Loops = High Utilisation, Happy Clients, Profitable Growth!